Then $(e_i, u_i, w_i) \geqslant (e', t, s')$ by the choice of $e \neq e'$. Hence $C' = C \cap X_{3-i}$, but also $A' = A \cup X_i$. Thus, any vertex or edge of $G[C]$ that is not in $G[C']$ lies in $G[X_i] \subseteq G[A']$. The fact that $\alpha_i(\sigma_t) \in \mathcal{T}$ thus implies $\alpha'_i(\sigma_t) \in \mathcal{T}$, as desired.

Let T be the tree obtained from the disjoint union of $T_1 - u_1$ and $T_2 - u_2$ by joining w_1 to w_2 by a new edge e. Let $\alpha\colon \vec{E}(T) \to \vec{S_k}$ map (e, w_{3-i}, w_i) to $\alpha'_i(e_i, u_i, w_i) = (X_i, X_{3-i})$ for $i = 1, 2$, and otherwise extend the α'_i. Then α commutes with the inversion of $\vec{e}$ and of (X_1, X_2), and $\alpha(\sigma_t) = \alpha'_i(\sigma_t) \in \mathcal{T}^* \cup \mathcal{T}^+$ for all $t \in T$, in particular for $t = w_i$. Hence, (T, α) satisfies (2). □

Theorem 12.5.9, as stated above, is a special case of a more general result in which $\mathcal{T}^*$ can be replaced by other collections $\mathcal{F}$ of 'forbidden' stars of oriented separations. Given a set S of separations, an *$\mathcal{F}$-tangle* of S is a consistent orientation of S that has no subset $\sigma \in \mathcal{F}$. For

$\mathcal{F}$-tangle

$$\mathcal{F}_k := \Big\{\sigma \subseteq \vec{S} : \sigma \text{ is a star and } \Big|\bigcap\{\, B : (A,B) \in \sigma\}\Big| < k\Big\}$$

we obtain the following more tangle-like duality theorem for tree-width:

Theorem 12.5.11. *The following two assertions are equivalent for all graphs G and integers $k > 0$:*

(i) *G has tree-width less than $k-1$;*

(ii) *G has no $\mathcal{F}_k$-tangle of S_k.*

It is not hard to show that (i) is equivalent to the existence of an S_k-tree over $\mathcal{F}_k$ in G (Ex. 59). Theorem 12.5.11 in that form and Theorem 12.5.9 are both corollaries of a more general theorem about tangle-tree duality in so-called 'abstract separation systems'; see the notes.

Moreover, any $\mathcal{F}_k$-tangle of S_k gives rise to a k-bramble (Ex. 59). Thus, Theorem 12.5.11 implies the tree-width duality theorem (12.4.3).

12.6 Tree-decompositions and forbidden minors

If $\mathcal{H}$ is any set or class of graphs, then the class

$$\mathrm{Forb}_{\preccurlyeq}(\mathcal{H}) := \{\, G \mid G \not\succcurlyeq H \text{ for all } H \in \mathcal{H}\,\}$$

$\mathrm{Forb}_{\preccurlyeq}(\mathcal{H})$

of all graphs without a minor in $\mathcal{H}$ is a graph property, i.e. is closed under isomorphism.[11] When it is written as above, we say that this property is expressed by specifying the graphs $H \in \mathcal{H}$ as *forbidden* (or *excluded*) *minors*.

forbidden minors

[11] As usual, we abbreviate $\mathrm{Forb}_{\preccurlyeq}(\{H\})$ to $\mathrm{Forb}_{\preccurlyeq}(H)$.

(1.7.1) By Proposition 1.7.1, $\mathrm{Forb}_{\preccurlyeq}(\mathcal{H})$ is closed under taking minors, or *minor-closed*: if $G' \preccurlyeq G \in \mathrm{Forb}_{\preccurlyeq}(\mathcal{H})$ then $G' \in \mathrm{Forb}_{\preccurlyeq}(\mathcal{H})$. Every minor-closed property can in turn be expressed by forbidden minors:

[5.2] **Lemma 12.6.1.** *A graph property $\mathcal{P}$ can be expressed by forbidden minors if and only if it is closed under taking minors.*

$\overline{\mathcal{P}}$ *Proof.* For the 'if' part, note that $\mathcal{P} = \mathrm{Forb}_{\preccurlyeq}(\overline{\mathcal{P}})$, where $\overline{\mathcal{P}}$ is the complement of $\mathcal{P}$. □

In Section 12.7, we shall return to the general question of how a given minor-closed property is best represented by forbidden minors. In this section, we begin by looking at a particular example of such a property: bounded tree-width.

Consider the property of having tree-width less than some given integer k. By Lemmas 12.4.1 and 12.6.1, this property can be expressed by forbidden minors. Choosing their set $\mathcal{H}$ as small as possible, we find that $\mathcal{H} = \{K^3\}$ for $k = 2$: the graphs of tree-width < 2 are precisely the forests. For $k = 3$, we have $\mathcal{H} = \{K^4\}$:

Proposition 12.6.2. *A graph has tree-width < 3 if and only if it has*
(7.3.1) *no K^4 minor.*

(12.3.2)
(12.3.5) *Proof.* By Corollary 12.3.5, we have $\mathrm{tw}(K^4) \geqslant 3$. By Lemma 12.4.1,
(12.3.6) therefore, a graph of tree-width < 3 cannot contain K^4 as a minor.
(12.4.1)

Conversely, let G be a graph without a K^4 minor; we assume that $|G| \geqslant 3$. Add edges to G until the graph G' obtained is edge-maximal without a K^4 minor. By Proposition 7.3.1, G' can be constructed recursively from triangles by pasting along K^2s. By induction on the number of recursion steps and Corollary 12.3.5, every graph constructible in this way has a tree-decomposition into triangles (as in the proof of Proposition 12.3.6). Such a tree-decomposition of G' has width 2, and by Lemma 12.3.2 it is also a tree-decomposition of G. □

As k grows, the list of forbidden minors characterizing the graphs of tree-width $< k$ seems to grow fast. They are known explicitly only up to $k = 4$; see the notes.

A question converse to the above is to ask for which H (other than K^3 and K^4) the tree-width of the graphs in $\mathrm{Forb}_{\preccurlyeq}(H)$ is bounded. This is the case, for example, when H is grid:

[12.7.1]
[12.7.3] **Theorem 12.6.3.** (Robertson & Seymour 1986)
For every integer r there is an integer k such that every graph of tree-width at least k has an $r \times r$ grid minor.

This *grid theorem* may, at first glance, look like just a specific and technical result. But it has a sweeping consequence:

Corollary 12.6.4. *Given a graph H, the graphs without an H minor have bounded tree-width if and only if H is planar.*

Proof. Since all grids and their minors are planar, every class $\mathrm{Forb}_{\preccurlyeq}(H)$ with a non-planar H contains all grids, which have unbounded tree-width (see after Theorem 12.4.3). (4.4.6)

Conversely, every planar graph H is a minor of some grid: take a drawing of the graph, fatten its vertices and edges, and superimpose a sufficiently fine plane grid. Hence, by Theorem 12.6.3, the graphs in $\mathrm{Forb}_{\preccurlyeq}(H)$ have bounded tree-width as soon as H is planar. □

Theorem 12.6.3 has another interesting application. Recall that a class $\mathcal{H}$ of graphs has the *Erdős-Pósa property* if the number of vertices in a graph needed to cover all its subgraphs in $\mathcal{H}$ is bounded by a function of its maximum number of disjoint subgraphs in $\mathcal{H}$. Now let H be a fixed connected graph, and consider the class $\mathcal{H} = IH$ of graphs that contract to a copy of H. (Thus, G has a subgraph in $\mathcal{H}$ if and only if $H \preccurlyeq G$.) H $\mathcal{H}$

Theorem 12.6.5. (Robertson & Seymour 1986)
$\mathcal{H}$ has the Erdős-Pósa property if H is planar.

Proof. We have to find a function $f\colon \mathbb{N} \to \mathbb{N}$ such that, given $k \in \mathbb{N}$ and a graph G, either G contains k disjoint models of H or there is a set U of at most $f(k)$ vertices in G such that $H \not\preccurlyeq G - U$. (12.3.1)

By Corollary 12.6.4, there exists for every $k \geqslant 1$ an integer w_k such that every graph of tree-width at least w_k contains the disjoint union of k copies of H (which is again planar) as a minor. Define

$$f(k) := 2f(k-1) + w_k$$

inductively, starting with $f(0) = f(1) = 0$.

To verify that f does what it should, we apply induction on k. For $k \leqslant 1$ there is nothing to show. Now let k and G be given for the induction step. If $\mathrm{tw}(G) \geqslant w_k$, we are home by definition of w_k. So assume that $\mathrm{tw}(G) < w_k$, and let $(T, (V_t)_{t\in T})$ be a tree-decomposition of G of width $< w_k$. Let us direct the edges t_1t_2 of the tree T as follows. Let T_1, T_2 be the components of $T - t_1t_2$ containing t_1 and t_2, respectively, and put

$$G_1 := G\Big[\bigcup_{t\in T_1} (V_t \smallsetminus V_{t_2})\Big] \quad \text{and} \quad G_2 := G\Big[\bigcup_{t\in T_2} (V_t \smallsetminus V_{t_1})\Big].$$

We direct the edge t_1t_2 towards G_i if $H \preccurlyeq G_i$, thereby giving t_1t_2 either one or both or neither direction.

If every edge of T receives at most one direction, we follow these to a node $t \in T$ such that no edge at t in T is directed away from t. As H is connected, this implies by Lemma 12.3.1 that V_t meets every $IH \subseteq G$. This completes the proof with $U = V_t$, since $|V_t| \leqslant w_k \leqslant f(k)$ by the choice of our tree-decomposition.

Suppose now that T has an edge t_1t_2 that received both directions. For each $i = 1, 2$ let us ask if we can cover all the models of H in G_i by at most $f(k-1)$ vertices. If we can, for both i, then by Lemma 12.3.1 the two covers combine with $V_{t_1} \cap V_{t_2}$ to the desired cover U for G. Suppose now that G_1 has no such cover. Then, by the induction hypothesis, G_1 contains $k-1$ disjoint models of H. Since t_1t_2 was also directed towards t_2, there is another such model in G_2. This gives the desired total of k disjoint models of H in G. □

Theorem 12.6.5 contains the Erdős-Pósa theorem 2.3.2 as the special case that $H = K^3$. It is best possible in that if H is non-planar, then $\mathcal{H} = IH$ does not have the Erdős-Pósa property (Exercise 62).

We conclude this section with some structure theorems for graphs not containing a given complete graph as a minor. These theorems are more difficult to prove than the results we have seen so far in this chapter, and they are not even that easy to state. But it's worth an effort: already the first of them, the excluded-K^n theorem, is both central to the proof of the graph minor theorem and can be applied elsewhere.

linear decom-position

A *linear decomposition* of G is a family $(V_i)_{i \in I}$ of vertex sets indexed by some linear order I such that $\bigcup_{i \in I} V_i = V(G)$, every edge of G has both its ends in some V_i, and $V_i \cap V_k \subseteq V_j$ whenever $i < j < k$. When G is finite, this is just a tree-decomposition whose decomposition tree is a path, and usually called a *path-decomposition*. If each V_i contains at most k vertices and k is minimal with this property, then $(V_i)_{i \in I}$ has *width* $k-1$.

$C_1, \dots, C_k$
$S - k$
cuffs

Let S' be a subspace of a surface[12] S obtained by removing the interiors of finitely many disjoint closed discs, with boundary circles $C_1, \dots, C_k$ say. This space is determined up to homeomorphism by S and the number k, and we denote it by $S - k$. Each C_i is the image of a continuous map $f_i \colon [0,1] \to S'$ that is injective except for $f_i(0) = f_i(1)$. We call $C_1, \dots, C_k$ the *cuffs* of S' and the points $f_1(0), \dots, f_k(0)$ their *roots*. The other points of each C_i are linearly ordered by f_i as images of $(0,1)$; when we use cuffs as index sets for linear decompositions below, we shall be referring to these linear orders. An *embedding* of a graph in S (or in $S - k$) is defined analogously to embeddings in the plane.

k-near embedding

Let H be a graph, S a surface, and $k \in \mathbb{N}$. We say that H is *k-nearly embeddable* in S if H has a set X of at most k vertices such that $H - X$ can be written as $H_0 \cup H_1 \cup \ldots \cup H_k$ so that

(N1) there exists an embedding $\sigma \colon H_0 \hookrightarrow S - k$ that maps only vertices to cuffs and no vertex to the root of a cuff;

[12] A compact connected 2-manifold without boundary; see Appendix B.

(N2) the graphs $H_1, \dots, H_k$ are pairwise disjoint (and may be empty), and $H_0 \cap H_i = \sigma^{-1}(\sigma(H_0) \cap C_i)$ for each i;

(N3) every H_i with $i \geqslant 1$ has a linear decomposition $(V_z^i)_{z \in C_i \cap \sigma(H_0)}$ of width $< k$ such that $\sigma^{-1}(z) \in V_z^i$ for all z.

Here, then, is the structure theorem for graphs without a K^n minor.[13] Note that, for $n = 5$, Wagner's Theorem (7.3.4) remains stronger and more precise. The case of $n = 4$ is covered by Proposition 7.3.1.

Theorem 12.6.6. (Robertson & Seymour 2003)
For every $n \geqslant 5$ there exists a $k \in \mathbb{N}$ such that every graph not containing K^n as a minor has a tree-decomposition whose torsos are k-nearly embeddable in a surface in which K^n is not embeddable.

Theorem 12.6.6 is true also for infinite graphs; see the notes.

Note that there are only finitely many surfaces in which K^n is not embeddable. The set of those surfaces in the statement of Theorem 12.6.6 could therefore be replaced by just two surfaces: the orientable and the non-orientable surface of maximum genus in this set.

Theorem 12.6.6 also has a converse, though only a qualitative one. A decomposition as described does not by itself preclude the presence of a K^n minor. But for every n there is an r such that no graph with such a decomposition has a K^r minor. This is because the adhesion sets of the tree-decomposition have bounded size, e.g. by $2k + n$, since they induce complete subgraphs in the torsos, and these are k-nearly embeddable in a surface that does not accommodate K^n.

For graphs without a given topological minor, there is a related structure theorem:

Theorem 12.6.7. (Grohe & Marx 2012)
For every $n \geqslant 5$ there exists a $k \in \mathbb{N}$ such that every graph not containing K^n as a topological minor has a tree-decomposition whose torsos are either k-nearly embeddable in a surface of Euler genus $\leqslant k$ or have at most k vertices of degree $> k$.

(See Appendix B for the definition of the Euler genus of a surface.)

There are also structure theorems for excluding infinite minors, and we now state two of these.

First, the structure theorem for excluding $K^{\aleph_0}$. Call a graph H *nearly planar* if H has a finite set X of vertices such that $H - X$ can be written as $H_0 \cup H_1$ so that (N1–2) hold with $S = S^2$ (the sphere) and $k = 1$, while (N3) holds with $k = |X|$. (In other words, deleting *nearly planar*

[13] Robertson and Seymour proved several versions of this theorem, of which Theorem 12.6.6 is the simplest. See the notes.

a bounded number of vertices makes H planar except for a subgraph of bounded linear width sewn on to the unique cuff of $S^2 - 1$.) A tree-decomposition $(T, (V_t)_{t \in T})$ of a graph G has *finite adhesion* if all its adhesion sets are finite and for every infinite path $t_1 t_2 \dots$ in T the value of $\liminf_{i \to \infty} |V_{t_i} \cap V_{t_{i+1}}|$ is finite.

finite adhesion

Unlike its counterpart for K^n, the excluded-$K^{\aleph_0}$ structure theorem has a direct converse. It thus characterizes the graphs without a $K^{\aleph_0}$ minor, as follows:

Theorem 12.6.8. *A graph G has no $K^{\aleph_0}$ minor if and only if G has a tree-decomposition of finite adhesion whose torsos are nearly planar.*

Finally, a structure theorem for excluding $K^{\aleph_0}$ as a topological minor. Let us say that G has *finite tree-width* if G admits a tree-decomposition $(T, (V_t)_{t \in T})$ into finite parts such that for every infinite path $t_1 t_2 \dots$ in T the set $\bigcup_{j \geqslant 1} \bigcap_{i \geqslant j} V_{t_i}$ is finite.

finite tree-width

Theorem 12.6.9. *The following assertions are equivalent for connected graphs G:*

(i) *G does not contain $K^{\aleph_0}$ as a topological minor;*

(ii) *G has finite tree-width;*

(iii) *G has a normal spanning tree T such that for every ray R in T there are only finitely many vertices v such that G contains an infinite v–$(R - v)$ fan.*

12.7 The graph minor theorem

Graph properties that are closed under taking minors occur frequently in graph theory. Among the most natural examples are the properties of being embeddable in some fixed surface, such as planarity.

By Kuratowski's theorem, planarity can be expressed by forbidding the minors K^5 and $K_{3,3}$. This is a *good characterization* of planarity in the following sense. Suppose we wish to persuade someone that a certain graph is planar: this is easy (at least intuitively) if we can produce a drawing of the graph. But how do we persuade someone that a graph is non-planar? By Kuratowski's theorem, there is also an easy way to do that: we just have to exhibit an IK^5 or $IK_{3,3}$ in our graph, as an easily checked 'certificate' for non-planarity. Our simple Proposition 12.6.2 is another example of a good characterization: if a graph has tree width < 3, we can prove this by exhibiting a suitable tree-decomposition; if not, we can produce an IK^4 as evidence.